

Performance Testing

Date	
Team ID	xxxxxx
Project Name	EduGenie: Google Gemini Powered Learning Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how EduGenie behaves under expected and peak load. Tests were run with the project's own asynchronous load-test script (EduGenie/tests/load_test.py).

Step 1: Testing Overview

Field	Details
Testing Tool Used	Custom Python load tester (tests/load_test.py, asyncio + httpx); pytest for functional tests; psutil for CPU/memory
Type of Testing	Load testing (10 concurrent users) and stress testing (50 concurrent users)
Target Module / API	GET /, GET /health, GET /qa, POST /explain, POST /quiz, POST /summarize, GET /learn/recommendations
Test Environment	Local - Uvicorn (1 worker) on a 4-core / 15.7 GB Linux machine. The Gemini call was replaced by an instant stub to measure EduGenie's own server overhead (end-to-end AI latency depends on Google's API and the key's quota).
Test Date	

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home page and health check load (GET /, GET /health)	10	10 each	Avg < 2 s, 0% errors
2	All five AI endpoints under normal load (Q&A, explain, quiz, summarize, learning path)	10	10 each	Avg < 2 s (server overhead), 0% errors
3	Stress test on the most-used endpoints (Q&A and quiz)	50	10 each	No failures, max < 5 s
4	End-to-end test with the live Gemini API (python tests/load_test.py --users 5 --duration 30)	5	30	AI responses within ~10 s; limited by the API key's rate limit

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	11-12 ms (10 users); 51-58 ms (50 users)	Pass	Server overhead only (Gemini stubbed)
2	Response Time (Max)	< 5 seconds	218 ms	Pass	Worst case across all scenarios
3	Throughput (Req/sec)	-	824-985 req/s	Pass	Single Uvicorn worker
4	Error Rate	< 1%	0.0%	Pass	~77,000 requests, no failures
5	CPU Utilization	< 80%	14.5% peak (of 4 cores)	Pass	Measured with psutil during load
6	Memory Utilization	< 80%	54 MB (0.3% of RAM)	Pass	Lightweight - no local model loaded
7	End-to-end AI response (live Gemini)	< 10 seconds			To be measured with a live API key (scenario 4)

Step 4: Observations & Analysis

- EduGenie's own processing (routing, validation, prompt building, JSON parsing, response) adds only ~12 ms on average, so total response time is dominated by the Gemini API.
- No errors occurred at 50 concurrent users; blocking Gemini calls run in FastAPI's thread pool, keeping the server responsive.
- Memory use stays very low (~54 MB) because AI inference runs in Google Cloud; enabling the optional local LaMini-Flan-T5 model adds roughly 3 GB.
- For production scale, run several Uvicorn workers or containers behind a load balancer; the stateless design makes horizontal scaling straightforward.
- Gemini overload (HTTP 503 'high demand') and free-tier rate limits (HTTP 429) are the most likely bottleneck under real load; EduGenie retries automatically with exponential backoff, falls back to backup models and otherwise shows a clear message with a Try again button.

Step 5: Screenshots / Evidence

```
(.venv) EduGenie> python tests/load_test.py --users 10 --duration 10 (Gemini stubbed)
EduGenie load test against http://127.0.0.1:8766
Scenario  Users  Requests  Avg ms  p95 ms  Max ms  Req/s  Errors
-----
home      10      8394      11.9    42.2    182.9   838.7   0.0%
health    10      8995      11.1    37.7    166.8   899.0   0.0%
qa         10      8871      11.3    38.6    143.4   886.5   0.0%
explain    10      8245      12.1    42.4    201.1   824.0   0.0%
quiz       10      8345      12.0    37.9    145.1   833.9   0.0%
summarize  10      8438      11.9    42.7    217.6   843.2   0.0%
path       10      8878      11.3    38.6    163.0   887.5   0.0%

(.venv) EduGenie> python tests/load_test.py --users 50 --duration 10 --scenarios qa,quiz
EduGenie load test against http://127.0.0.1:8766
Scenario  Users  Requests  Avg ms  p95 ms  Max ms  Req/s  Errors
-----
qa         50     9872      50.6    73.0    107.8   984.8   0.0%
quiz       50     8659      57.8    70.3    104.0   863.2   0.0%

Server resources during load: cpu_avg=10.1% cpu_max=14.5% (of all 4 cores; single-core max 58%) mem_max=54MB (0.3% of 15.7GB)
```

Load test output (tests/load_test.py) with server resource usage

```
(.venv) EduGenie> pytest -v
===== test session starts =====
platform linux -- Python 3.11.15, pytest-9.1.1, pluggy-1.6.0 -- /home/user/EduGenie-AI/.venv/bin/python
cachedir: .pytest_cache
rootdir: /home/user/EduGenie-AI/EduGenie
configfile: pytest.ini
testpaths: tests
plugins: anyio-4.15.1
collecting ... collected 43 items

tests/test_api.py::test_index_page_renders PASSED [ 2%]
tests/test_api.py::test_static_files_served PASSED [ 4%]
tests/test_api.py::test_health PASSED [ 6%]
tests/test_api.py::test_qa PASSED [ 9%]
tests/test_api.py::test_qa_requires_question PASSED [ 11%]
tests/test_api.py::test_explain PASSED [ 13%]
tests/test_api.py::test_explain_trailing_slash_like_original_doc PASSED [ 16%]
tests/test_api.py::test_explain_empty_topic PASSED [ 18%]
tests/test_api.py::test_explain_topic_too_long PASSED [ 20%]
tests/test_api.py::test_summarize PASSED [ 23%]
tests/test_api.py::test_summarize_empty PASSED [ 25%]
tests/test_api.py::test_quiz PASSED [ 27%]
tests/test_api.py::test_quiz_num_questions PASSED [ 30%]
tests/test_api.py::test_quiz_invalid_num_questions PASSED [ 32%]
tests/test_api.py::test_quiz_invalid_ai_output PASSED [ 34%]
tests/test_api.py::test_quiz_empty PASSED [ 37%]
tests/test_api.py::test_learning_recommendations PASSED [ 39%]
tests/test_api.py::test_learning_recommendations_requires_topic PASSED [ 41%]
tests/test_api.py::test_gemini_failure_returns_502 PASSED [ 44%]
tests/test_api.py::test_missing_api_key_returns_503 PASSED [ 46%]
tests/test_gemini_retry.py::test_retries_busy_model_then_succeeds PASSED [ 48%]
tests/test_gemini_retry.py::test_falls_back_to_next_model_when_primary_stays_busy PASSED [ 51%]
tests/test_gemini_retry.py::test_rate_limit_is_retried PASSED [ 53%]
tests/test_gemini_retry.py::test_all_models_busy_raises_busy_error PASSED [ 55%]
tests/test_gemini_retry.py::test_invalid_key_is_not_retried PASSED [ 58%]
tests/test_gemini_retry.py::test_unknown_model_moves_to_fallback_without_retry PASSED [ 60%]
tests/test_gemini_retry.py::test_fallback_can_be_disabled PASSED [ 62%]
tests/test_gemini_retry.py::test_fallback_list_skips_primary_and_duplicates PASSED [ 65%]
tests/test_gemini_retry.py::test_busy_error_returns_503_with_retry_after PASSED [ 67%]
tests/test_live_gemini.py::test_live_qa SKIPPED (set RUN LIVE TESTS=...) [ 69%]
tests/test_live_gemini.py::test_live_quiz SKIPPED (set RUN LIVE TESTS=...) [ 72%]
tests/test_live_gemini.py::test_live_learning_path SKIPPED (set RUN LIVE TESTS=...) [ 74%]
tests/test_modules.py::test_clean_json_block_removes_fences PASSED [ 76%]
tests/test_modules.py::test_parse_quiz_accepts_wrapped_object_and_skips_bad_items PASSED [ 79%]
tests/test_modules.py::test_parse_quiz_extract_array_from_surrounding_text PASSED [ 81%]
tests/test_modules.py::test_parse_quiz_raises_on_garbage PASSED [ 83%]
tests/test_modules.py::test_quiz_format_error_is_gemini_error PASSED [ 86%]
tests/test_modules.py::test_learning_path_invalid_level_defaults_to_beginner PASSED [ 88%]
tests/test_modules.py::test_explain_falls_back_to_gemini_when_local_fails PASSED [ 90%]
tests/test_modules.py::test_explain_uses_local_model_when_selected PASSED [ 93%]
tests/test_modules.py::test_generate_without_key_raises PASSED [ 95%]
tests/test_modules.py::test_placeholder_key_is_treated_as_missing PASSED [ 97%]
tests/test_modules.py::test_model_name_default_and_override PASSED [100%]

===== 40 passed, 3 skipped in 0.49s =====
```

Functional test suite: 40 passed, 3 live tests skipped (require API key)